

Laxatives and anti-diarrhoeal agents



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Objectives



- List the types of laxatives.
- Know the mechanism of action of laxatives.
- Describe anti-diarrhoeal agents with their mechanism of action.

Constipation



- A practical definition of constipation, used by many patients, is simply a reduced frequency or ease of stool passage from what is regarded as normal or expected pattern for that individual.

Treatment of constipation



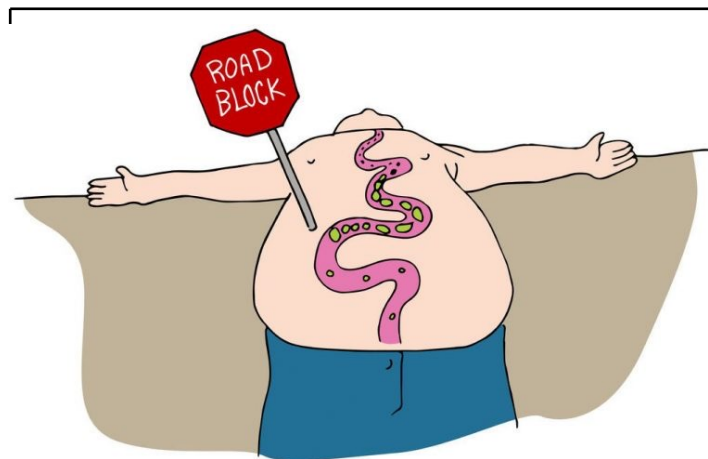
Non-drug therapy: It is the first line in all cases of constipation

- Diet rich in fibers e.g. fruits, vegetables, whole meal bread, etc.
- Increase fluid intake
- Minimize tea and coffee.
- Physical exercise to activate abdominal muscles and intestinal peristalsis. This helps food move more efficiently through the gut.

Drug therapy(Laxatives)



- Laxatives are commonly used for constipation to accelerate the movement of food through the GI tract.



- Laxatives increase the potential for loss of pharmacologic effect of poorly absorbed, delayed-acting, and extended-release oral preparations by accelerating their transit through the intestines.
- They may also cause electrolyte imbalances when used chronically.
- Many of these drugs have a risk of dependency for the user.

General indications of laxatives

- **Constipation:** laxatives should not be used for prolonged duration to avoid laxative dependence.
- To increase excretion of toxic substances from the GIT.
- To prepare the bowel before X-ray or colonoscopy.
- Painful anal conditions e.g. anal fissure or piles.
- Postoperative: e.g. after hemorrhoids (piles) to avoid strain.

Contraindications of laxatives

- Laxatives are dangerous in cases of undiagnosed abdominal pain or inflammatory bowel disease. They may lead to intestinal perforation.
- Organic obstruction of the GIT.

Classification of laxatives according to mechanism of action

IRRITANTS and STIMULANTS

Bisacodyl CORRECTOL, DULCOLAX

Castor oil GENERIC ONLY

Senna EX-LAX, SENOKOT

BULK LAXATIVES

Methylcellulose CITRUCEL

Psyllium METAMUCIL

SALINE and OSMOTIC LAXATIVES

Lactulose CONSTULOSE, ENULOSE

Magnesium citrate CITROMA

Magnesium hydroxide MILK OF MAGNESIA

Polyethylene glycol GOLYTELY, MIRALAX

STOOL SOFTENERS

Docusate COLACE

LUBRICANT LAXATIVES

Glycerin suppositories GENERIC ONLY

Mineral oil GENERIC ONLY

A-Irritants and stimulants



1. **Senna:** This agent is a widely used stimulant laxative.
 - Taken orally, *Senna* causes evacuation of the bowels within 8 to 10 hours.
 - It also causes water and electrolyte secretion into the bowel.



2. Bisacodyl: Available as suppositories and enteric-coated tablets, *bisacodyl* is a potent stimulant of the colon.

- It acts directly on nerve fibers in the mucosa of the colon.

- 3. Castor oil:** This agent is broken down in the small intestine to ricinoleic acid, which is a very irritating substance and promptly increases peristalsis.
- Pregnant patients should avoid *castor oil* because it may stimulate uterine contractions.

B-Bulk laxatives



- The bulk laxatives include hydrophilic colloids (from indigestible parts of fruits and vegetables).
- They form gels in the large intestine, causing water retention and intestinal distension, thereby increasing peristaltic activity.

- Similar actions are produced by *methylcellulose*, psyllium seeds, and bran.
- They should be used cautiously in patients who are immobile because of their potential for causing intestinal obstruction.



C. Saline and osmotic laxatives



- Saline cathartics, such as *magnesium citrate* and *magnesium hydroxide*, are nonabsorbable salts that hold water in the intestine by osmosis.
- This distends the bowel, increasing intestinal activity and producing defecation in a few hours.
- Electrolyte solutions containing *polyethylene glycol* (PEG) are used as colonic lavage solutions to prepare the gut for radiologic or endoscopic procedures.

- *Lactulose* is a semisynthetic disaccharide sugar that acts as an osmotic laxative.
- It cannot be hydrolyzed by GI enzymes but is degraded by colonic bacteria into lactic, formic, and acetic acids.
- This increases osmotic pressure, causing fluid accumulation, colon distension, soft stools, and defecation.

D. Stool softeners (emollient laxatives or surfactants)



- Surface-active agents that become emulsified with the stool produce softer feces and ease passage. These include *docusate sodium* and *docusate calcium*.
- They may take days to become effective and are often used for prophylaxis rather than acute treatment.

E. Lubricant laxatives



- *Mineral oil* and *glycerin suppositories* are lubricants and act by facilitating the passage of hard stools.
- *Mineral oil* should be taken orally in an upright position to avoid its aspiration and potential for lipid pneumonia.

F. Chloride channel activators



- *Lubiprostone*, currently the only agent in this class, works by activating chloride channels to increase fluid secretion in the intestinal lumen.
- This eases the passage of stools and causes little change in electrolyte balance.

- *Lubiprostone* is used in the treatment of chronic constipation, particularly because tolerance or dependency has not been associated with this drug.
- Also, drug– drug interactions appear minimal because metabolism occurs quickly in the stomach and jejunum.

Anti-diarrhoeal agents



Increased motility of the GI tract and decreased absorption of fluid are major factors in diarrhoea.

There are 3 approaches to the treatment of severe acute diarrhoea:

- 1-maintenance of fluid and electrolyte balance
- 2-use of anti-infective agents
- 3-use of spasmolytic or other antidiarrhoeal agents

Classification

ANTIMOTILITY AGENTS

Diphenoxylate + atropine LOMOTIL

Loperamide IMODIUM A-D

ADSORBENTS

Aluminum hydroxide GENERIC ONLY

Methylcellulose CITRUCEL

AGENTS THAT MODIFY FLUID AND ELECTROLYTE TRANSPORT

Bismuth subsalicylate PEPTO-BISMOL

A. Antimotility agents



- Two drugs that are widely used to control diarrhoea are *diphenoxylate* and *loperamide*.
- Both are analogs of *meperidine* and have opioid-like actions on the gut.
- They activate presynaptic opioid receptors in the enteric nervous system to inhibit acetylcholine release and decrease peristalsis.

- At the usual doses, they lack analgesic effects.
- Because these drugs can contribute to toxic megacolon, they should not be used in young children or in patients with severe colitis.

B. Adsorbents



- Adsorbent agents, such as *aluminum hydroxide*, *activated charcoal* and *methylcellulose*, are used to control diarrhoea.
- Presumably, these agents act by adsorbing intestinal toxins or microorganisms and/or by coating or protecting the intestinal mucosa.
- They are much less effective than ant motility agents, and they can interfere with the absorption of other drugs.

C. Agents that modify fluid and electrolyte transport



- *Bismuth subsalicylate*, used for traveler's diarrhoea, decreases fluid secretion in the bowel.
- Its action may be due to its salicylate component as well as its coating action.
- Adverse effects may include black tongue and black stools.

References



- **Lippincott's Illustrated Reviews Pharmacology 7th edition.**
- **Rang & Dale's Pharmacology 8th edition.**